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Venture: AeroFlora Systems **Generated:** 2 Nov 2026

ASSIGNMENT 2 — VENTURE APP BUILD BRIEF

VENTURE APPLICATION OBJECTIVE

Build an individual working application prototype based on your validated venture logic. This document defines the exact software specifications derived from your Venture Quest journey.

Venture Name
AeroFlora Systems

Target User
Independent, off-grid organic tree-nut and specialty crop growers in Southern Arizona managing 20–100 acres, who currently spend over $4,000 monthly on trucked water during dry season peaks. Customer Profile Card AeroFlora Systems — Targeting high-value, off-grid Southwestern agriculture.

Core Problem
Global freshwater scarcity affects over two billion people, but traditional atmospheric water generators fail in low-humidity environments, consume massive amounts of grid electricity, and are cost-prohibitive for off-grid communities.

Jobs to Be Done (JTBD)
Extended micro-droughts drop local water tables below pump level and ambient humidity sits between 15%–30% during peak growing seasons, -> Extract daily operational freshwater directly on-site using energy-passive atmospheric harvesting, -> Maintain crop yield stability and protect my business margin without relying on high-cost water trucking or expensive diesel generators.

Value Proposition
For Independent, off-grid organic tree-nut and specialty crop growers in Southern Arizona managing 20–100 acres, who currently spend over $4,000 monthly on trucked water during dry season peaks. Customer Profile Card AeroFlora Systems — Targeting high-value, off-grid Southwestern agriculture., who struggles with Extreme crop loss risks, skyrocketing operational expenses from water-hauling trucks ($3,000–$8,000/month), and groundwater depletion during hyper-arid drought periods., our venture provides [product/service] that delivers [main benefit], so that [desired outcome].

Core MVP Features (Must Have)
1. Biomimetic Passive Condensation & Solar Mineralization Module: This is the core technical differentiator. Without verified water extraction at 15%–25% RH, the venture has no functional value proposition over existing alternatives. (Solves: On-site water generation in extreme low-humidity environments without grid electricity connection.)

What Will NOT Be Included in Version 1
Integrated automated crop-scheduling AI software, multi-tower synchronized mesh networking, custom aesthetic casing, and ultra-high-capacity multi-thousand gallon storage vaults will be explicitly excluded from Version 1 to prevent development delays and CapEx bloat.

Required Inputs from User
User credentials, business input variables, customer preferences, activity logs

Required Outputs / Display
Decision recommendations, calculated metrics, visual feedback, downloadable deliverables

Data That Must Be Stored
User profile, venture entities, session inputs, timestamps, status flags

Suggested User Flow
1. Onboarding & problem framing
2. User decision / calculation input
3. Value generation / result display
4. Save, export, or action completion

Validation Objective
The core biomimetic atmospheric water harvesting tower can reliably extract at least 450 liters of pure water daily in hyper-arid conditions (15%–25% relative humidity) on an off-grid farm site without fouling or performance drop.

SUBMITTED WORKING APP URL

https://aeroflora-systems.ai.studio/